

Ms. Neerah H. Ahuja | A.Y.: 2025-26 | Sem./Year: 5 | Course: 2302CS501 – Mobile Programming using Flutter | Slot Type: Lab (TA) | Division: CE-5A

Sr.	Course Content	Planning Title	Planning Description
1	Introduction to Dart Programming Language	Practice Dart Program	1) W.A.P. to print your name in console. (A) 2) W.A.P. to find percentage of 5 subject. (A) 3) Body Mass Index (BMI) is measure of health on weight. It can be calculated by taking your weight in kilograms and dividing by square of your height in meters. Write a program that prompts the user to enter weight in pounds and height in inches and display the BMI. Note: 1 pound=.45359237 Kg and 1 inch=0.254 meters. (A) 4) W.A.P. to check whether the given number is positive or negative. (A) 5) W.A.P. to perform Addition, Subtraction, Multiplication, Division based on user choice using if, if..else..if, & switch. (B)
2	Introduction to Dart Programming Language	Practice Program using Array and List	1) W.A.P. to count number of even or odd number from an array of n numbers. (A) 2) W.A.P. to insert int numbers into List and display in terminal. (A) 3) W.A.P. to insert Strings into List and display in terminal. (A) 4) W.A.P. to insert double into List and display in terminal and do sum of all numbers and print it with 2 decimal only. Ex: Total = 22.39998 => 22.40 (A) 5) W.A.P. to sort values in List. Ex. Input : [13, 2, -11, 142, -389, 0] Result: [-389, -11, 0, 2, 13, 142] (B)
3	Introduction to Dart Programming Language	Practice Program using Class and Object	1) Create a class named Candidate with Candidate_ID, Candidate_Name, Candidate_Age, Candidate_Weight and Candidate_Height data members. Also create a method GetCandidateDetails() and DisplayCandidateDetails(). Create main method to demonstrate the Candidate class. (A) 2) W.A.P. to create class Animal that has a method called animalSound() Subclass of Animals is cat it has its own implementation of an animal sound. (B) 3) W.A.P. to find friends detail by their name using dictionary. (Create local dictionary and search from it using Map & Model Class) (C)
4	Flutter Architecture & UI Design	Write a Flutter Code using Container Widget and Expanded Widget	1) Create a flutter project with proper name in Android Studio and observe different files for programming. (A) 2) W.A.P. to divide horizontal space of the screen into 3 different equal parts with different colors. (A) 3) W.A.P. to divide the vertical space of the screen into 3 different equal parts with different colors. (A)

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			4) W.A.P. to divide the whole screen into 9 (equal size and different size) with different colors. (B)
5	Flutter Architecture & UI Design	Write a Flutter Code using Text Widget and TextField Widget using Validation	1) W.A.P. to display "hello world" using Text widget. Change color & size of text using different properties. (A) 2) W.A.P. to use TextField and print the input value into the terminal using the controller. (A) 3) W.A.P. to do validation in the login screen. (Email Validation & Password Validation) on Button click. (B)
6	Flutter Architecture & UI Design	Write a Flutter Code for Use Different Widgets	1) W.A.P. to Add password visibility icon in the Text field. (A) 2) W.A.P. to create a registration screen using different widgets (Text, TextFromField, SizedBox, Radio, CheckBox, List, Button etc...)(B)
7	Flutter Architecture & UI Design	Write a Flutter Code using Stack Widget, Image Widget	1) W.A.P. to display an Image into the Image asset widget from the asset folder. (A) 2) W.A.P. to use Stack widget and show text upon an Image. (A) 3) W.A.P. to create a birthday card using different widgets.(A) 4) W.A.P. to display an Image from a WEB URL using a cached network Image. (B)
8	Flutter Architecture & UI Design	Design a Flutter UI using Different Widgets-Part-1	1) W.A.P. to Design simple calculator UI using Column, TextField, ElevatedButton, and Row. (A) 2) W.A.P. to Design Shopping Item Card using Card, Column, Image, Text, and ElevatedButton. (A) 3) W.A.P. to Design Simple Survey Form using Column, TextField, RadioListTile, CheckBox, and ElevatedButton. (B)
9	Flutter Architecture & UI Design	Design a Flutter UI using Different Widgets-Part-2	1) W.A.P. to Design Feedback Form UI using Column, TextFormField, ElevatedButton, and SnackBar. (A) 2) W.A.P. to Build a Quote Generator App using Column, Text, ElevatedButton, and SnackBar (B)
10	Flutter Architecture & UI Design	Design a Flutter UI using Different Widgets-Part-3	1) W.A.P. to Design To-Do List UI using Column, Text, Checkbox, ElevatedButton, and Padding. (A) 2) W.A.P. to Design Music Player UI using Card, Text and Buttons. (B)
11	Scrollable Widgets & Navigation	Write a Flutter Code using ListView and GridView	1) W.A.P. to create list of numbers and display using ListView. (A) 2) W.A.P. to create list of numbers and display using ListView.builder. (A) 3) W.A.P. to create list of images path in List and display it using GridView and then GridView.builder. (B)

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12	Scrollable Widgets & Navigation	Write a Flutter Code using ListView and ListView.builder	1) W.A.P. Design Contact List UI use ListView to display a list of contact with CircleAvatar, Text, and Trailling Icon. (A) 2) W.A.P. Design E-commerce UI use ListView.builder to display a list of items with CircleAvatar, Text, and ElevatedButton. (B)
13	Scrollable Widgets & Navigation	Write a Flutter Code using GridView and GridView.builder	1) W.A.P. Design Movie Poster UI using GridView, Card, Image. (A) 2) W.A.P.. Design Photo Gallery UI using GridView.builer, Card, Image and InkWell. (B)
14	Scrollable Widgets & Navigation	Write a Flutter Code using PageView and PageView.builder	1) W.A.P. Design PageView with 3D Page Transition Effect. (A) 2) W.A.P. Design Quotes Swiper UI using PageView, Card, Text and IconButton.(B)
15	Scrollable Widgets & Navigation	Write a Flutter Code using TabBar and Navigation Drawer	1) W.A.P. to use TabView and display different pages on different tab clicks. (A) 2) W.A.P. to use the NavigationDrawer & display different pages on different menu clicks. (B)
16	Scrollable Widgets & Navigation	Write a Flutter Code for Navigation Between Pages	1) W.A.P. to navigate from one page to another page using material page route. (A) 2) W.A.P. to navigate from one page to another page and pass data to another page using constructor. (B)
17	Scrollable Widgets & Navigation	Build Student Management App using SQLite Database	1) Create Database using Helper Class. (A) 2) Create Student Model Class. (B)
18	Introduction to Database & Dialog Control	Build Student Management App using SQLite Database-Part-1	1) Design Login Screen and Fetch Username and Password from Database. (A) 2) Design Add Student Screen (Name, Enrollment Number, Semester, Branch , Mobile Number) (B)
19	Introduction to Database & Dialog Control	Build Student Management App using SQLite Database-Part-2	1) Design Home Screen (List Students) with Edit and Delete Button. (A) 2) Design Student Detail Screen when click on Student Name All Details of particular Student Open in New Screen. (B)

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20	Introduction to Database & Dialog Control	Build Student Management App using SQLite Database-Part-3	1) Create Edit Student Screen. (A) 2) Create Delete Student Functionality. (B)
21	Introduction to Database & Dialog Control	Write a Flutter Code for Alert Dialog, Date Picker and JSON File	1) W.A.P. to create Alert Dialog with material & Cupertino theme and display it with 1“Ok” button. (A) 2) W.A.P. to generate model class from json. (A) 3) W.A.P. to fetch json file from asset & parse json data and display data in widget. (A) 4) W.A.P. to show DatePickerDialog on button click and set selected date. (B)
22	Accessing Rest API	Write a Flutter Code using Rest API-Part-1	1) W.A.P. to call rest API (using Mockapi) with GET Method with HTTP client library and display data in List widget using future builder. (A) 2) W.A.P. to pass Path Variable and Query Variable as parameters in GET Method and display the single record in the full- screen page. (B)
23	Accessing Rest API	Write a Flutter Code using Rest API-Part-2	1) W.A.P. to call rest API and display data in ListView.builder with pagination (Lazy Loading). (A)
24	Accessing Rest API	Write a Flutter Code using Rest API-Part-3	1) W.A.P. to design entry screen with Name, DOB, City, Address, etc. validate it & save data to webserver using POST method with form-urlencoded. (A)
25	Accessing Rest API	Write a Flutter Code using Rest API-Part-4	1) W.A.P. to display Country List in dropdown on selection of country call state API and fill data in state list after selection state call city API and fill city list in dropdown. (A)
26	Accessing Rest API	Write a Flutter Code using Rest API-Part-5	1) W.A.P. to Design Product Listing UI Fetch Product Data from API. (A)
27	Accessing Rest API	Write a Flutter Code using Rest API-Part-6	1) W.A.P. to Design News app rest API and display news in GridView.builder with pagination (Lazy Loading). (A)
28	Accessing Rest API	Write a Flutter Code using Rest API-Part-7	1) W.A.P. to design Contact US form with Name, Email, Message validate it & save data to webserver using POST method with form-urlencoded. (A)
29	Accessing Rest API	Write a Flutter Code using Rest API-Part-8	1) W.A.P. to display a list of movies in a dropdown. On selecting a movie, fetch its details, and display the cast & reviews dynamically using TMDb API. (A)

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30	Accessing Rest API	Write a Flutter Code using Rest API-Part-9	1) W.A.P. Design and create your own fake API for food and fetch the data into the Flutter UI. (A)
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